
Cloning of ds-stappled peptides for expression in *E. coli*

Introduction

This was developed as part of the protocol for screening of disulphide stapled peptides tagged with N-termini VnP15 - 6xH - Sumo tags at low scale, yielding approximately 100 µM protein solutions. This is based on the In vivo assembly (IVA) cloning [1]

The map of the construct insert designed for this protocol is: <https://benchling.com/s/seq-iZkZPeMI1QTVkMLIGTxX?m=slm-xdpvMtG07RHw0d7I54oA>

Materials

› PCR assembly

- › Q5 Master mix (NEB Cat No. M0492): 5 µl per reaction.
- › 1 ng /µl of reaction template DNA (purified plasmid in water or EB buffer)
- › 0.5 µM of each reverse and forward assembly primer
- › Eva Green dye
- › DNase free water

› Clean-up:

- › DpnI (NEB, Cat. No. R0176S): 0.25 µl per reaction

› Transformation:

- › 15 µl of BL21 DE3 competent cell (NEB, Cat No. C2527) per reaction

› Consumables:

- › 1 x MicroAmp™ EnduraPlate™ Optical 96-Well Clear Reaction Plates per 96 reactions (ThermoFisher, Cat.No. A36924)
- › 1 x qPCR compatible plate (For example: BioRad Cat. No. HSL9645) for 96 reactions.
- › 1 x 96-deep wells plate (round bottom, sterile, 2 ml).

› Equipment:

- › qPCR machine
- › Water bath (42°C)
- › Plate shaker incubator at 37°C

Procedure

PCR assembly:

1. Prepare the primers: the primers are shipped freeze-dried by IDT and need to be eluted in nuclease free water to a 2.5 - 5 uM concentration for their use in the assembly reaction. To resuspend the primers:

1. Centrifuge the plate for 2-3 min at 2,000 x g to make sure that the primer is in the bottom of the plate.
2. Add nuclease-free water to dissolve the primer (100 µL of water or TE buffer for 5 uM primer solution when receiving 0.5 nmol of primer) . Use the following [calculator](#) if different amounts/concentrations are used:
3. Mix by pipetting and incubate in shaker for 15 min to make sure it is completely dissolved.
4. Centrifuge the plate briefly before removing the film

2. Prepare a qPCR reaction mix calculation sheet by adjusting the values presented in the template in Table 1. Change the value in Reaction number ([Blue](#)) to adjust the number of reaction to be included. Change the reaction volume ([Orange](#)) if you plan to change the scale of the reaction. Adjust the value for the concentration of plasmid used as template ([Green](#)) and the primer concentration ([Violet](#)). [**Note:** Remember to always add an excess of reactions, especially if doing multiple reactions at once. A 3-5% excess is usually ok]

Table1. PCR mix template



	A	B	C	D	E	F
1	PCR SETUP					
2	Component	Units	Initial Conc	Final Conc	Vol (rx)	Vol (MM)
3	Q5 MM	X	2.0	1.0	5.00	495.00
4	FW Primer	µM	5.0	0.5	1.00	99.00
5	RV Primer	µM	5.0	0.5	1.00	99.00
6	Template DNA	ng/ul	40.0	0.8	0.20	19.80
7	Eva Green	X	20.0	1.0	0.50	49.50
8	Water	X	1.0	1.0	2.30	227.70
9	Reaction No.				99.00	792.00
10	Total Rx Vol				10.00	

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3. Prepare the PCR mix by mixing water, plasmid (template DNA), dye (Eva Green), Enzyme mix (Q5). Mix the solution by pipetting very slowly with the p1000 and by inversion - slowly 6-8 times.
 4. On a 96-well qPCR plate, dispense the forward and reverse primers at the bottom of the well (change tips between).
 5. Add the qPCR mix and mix by pipetting slowly 2-3 times.
 6. Set the reaction in CFX qPCR machine using the following heating programme:

The template below is set to work with cloning primers with 3' terminations corresponding to fragments: YFB001F and YFB001R. If different primers are used, adjust the temperature accordingly. Similarly if a different plasmid is used adjust the length of extension.

The number of cycles was optimised to 20. However, you may adjust this to work for you.

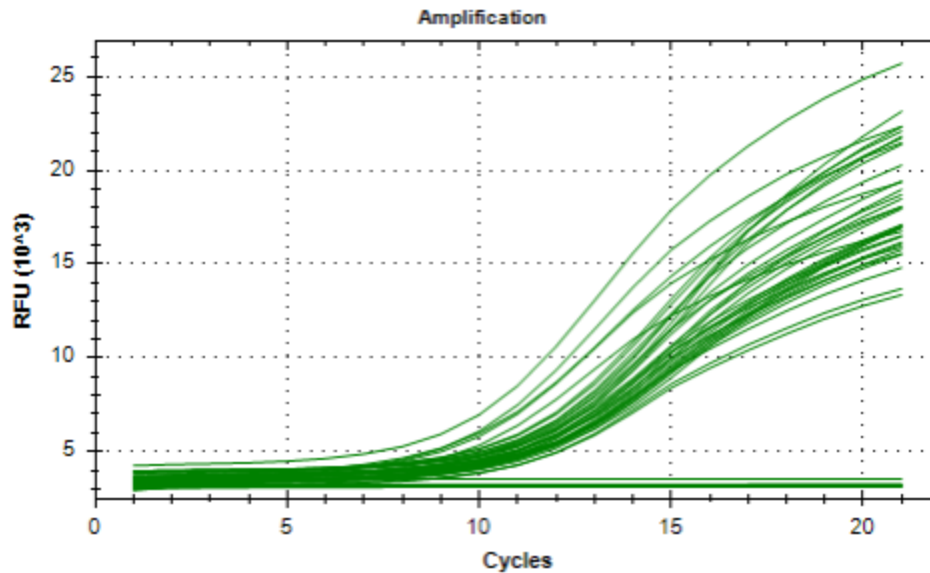
Table 2. Assembly PCR programme



	A	B	C	D	E	F
1	Heating Programme					
2	Stage		Temp (oC)	Time	Cycles	
3	Initial Denaturation		98	30	1	
4	Denaturation		98	10'	20	
5	Annealing/Extension		72	140'		
6	Final extension		72	2"	1	

7. Verify the efficiency of the amplification

CRITICAL Check that the exponential phase started before or at latest cycle 15 and that the amplification yielded a signal $> 5 \times 10^3$ RFU. This is essential for having high transformation efficiency. See for example (flat lines are negative controls):



DpnI digestion:

8. Removal of the template plasmid DNA is done with 4-5 Units of DpnI, which cleaved methylated DNA. Dilute DpnI (20 units/ μ l) 1:5 in EB buffer (10 mM Tris-HCL, pH8). Add 1 μ l of the DpnI dilution into each of the wells containing the amplified product and incubate at 37°C for 45 min, after this heat-inactivate the enzyme at 80°C for 20 min.
9. Cool the reaction product on ice and spin briefly before opening to avoid aerosols.

Transformation:

10. **CRITICAL** **Before starting:** make sure you have cooled p20 tips (in the freezer) and a cooling 96 well rack and a microAmp plate at 4°C on ice. Alternatively, if using a repeating pipette, make sure that the tips are cooled. [Here is presented the protocol for use with single tips, but you can choose to do it with a repeater pipette].
11. Thaw 8 tubes of NEB BL21 DE3 cells per plate of PCR reactions for approx 15 min (if using 200 μ l aliquots).
12. On a microAmp plate, using a p10 multichannel pipette dispense 1 μ l of the amplified DNA product at the bottom of each corresponding well (e.g. A1 > A1; C5>C5).

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13. Tick the cell aliquot 3–4 times lightly with your fingers to resuspend the cells (do not pipette mix!). Dispense 15 µl of cells per µl of DNA template used, on the bottom of each well. Change tips in between.
 14. Alternatively, using the repeater pipette add the cells to the bottom of the well before adding the DNA, then add the DNA and mix by swirling the tip around the well.
 15. Incubate the Cells and DNA for exactly 30 min (not less than 25) on ice without moving.
 16. Heat-shock the cells at exactly 42°C for 20 seconds - on a water bath.
 17. Put the cells on ice for 5 min
 18. Resuspend the cells in 100 µl of SOC media and cover the plate with a breathable film.
 19. Let the cell to recover for exactly 1h at 37°C in a plate shaker incubator.
 20. Dispense recovered cells on 900 µl of freshly made Kanamycin (50 µg/ml) selective LB media (if using YFB plasmids) in a deep well plate and grow overnight (maximum of 16 h) at 37°C, shaking.

References:

21. García-Nafria, J., Watson, J. & Greger, I. IVA cloning: A single-tube universal cloning system exploiting bacterial *In Vivo* Assembly. *Sci Rep* **6**, 27459 (2016). <https://doi.org/10.1038/srep27459>.